

DESIGN AND OPTIMIZATION OF HYBRID ELF-RESLT ARCHITECTURES FOR LONG-TAILED RECOGNITION



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WORK DONE

- Implemented and the “*ResLT: Residual Learning for Long-tailed Recognition*” paper from scratch.
- Designed and experimented with hybrid architectures combining “*ELF: An Early-Exiting Framework for Long-Tailed Classification*” and “*ResLT: Residual Learning for Long-tailed Recognition*” .
- Iteratively debugged, analyzed, and refined the models to achieve state-of-the-art performance.

FOUNDATIONAL CONCEPTS

ResLT's Contribution: Class-Level Re-balancing

- Introduced re-balancing in the parameter space.
- Its specialized three-branch head is explicitly designed to improve performance on under-represented medium and tail classes.

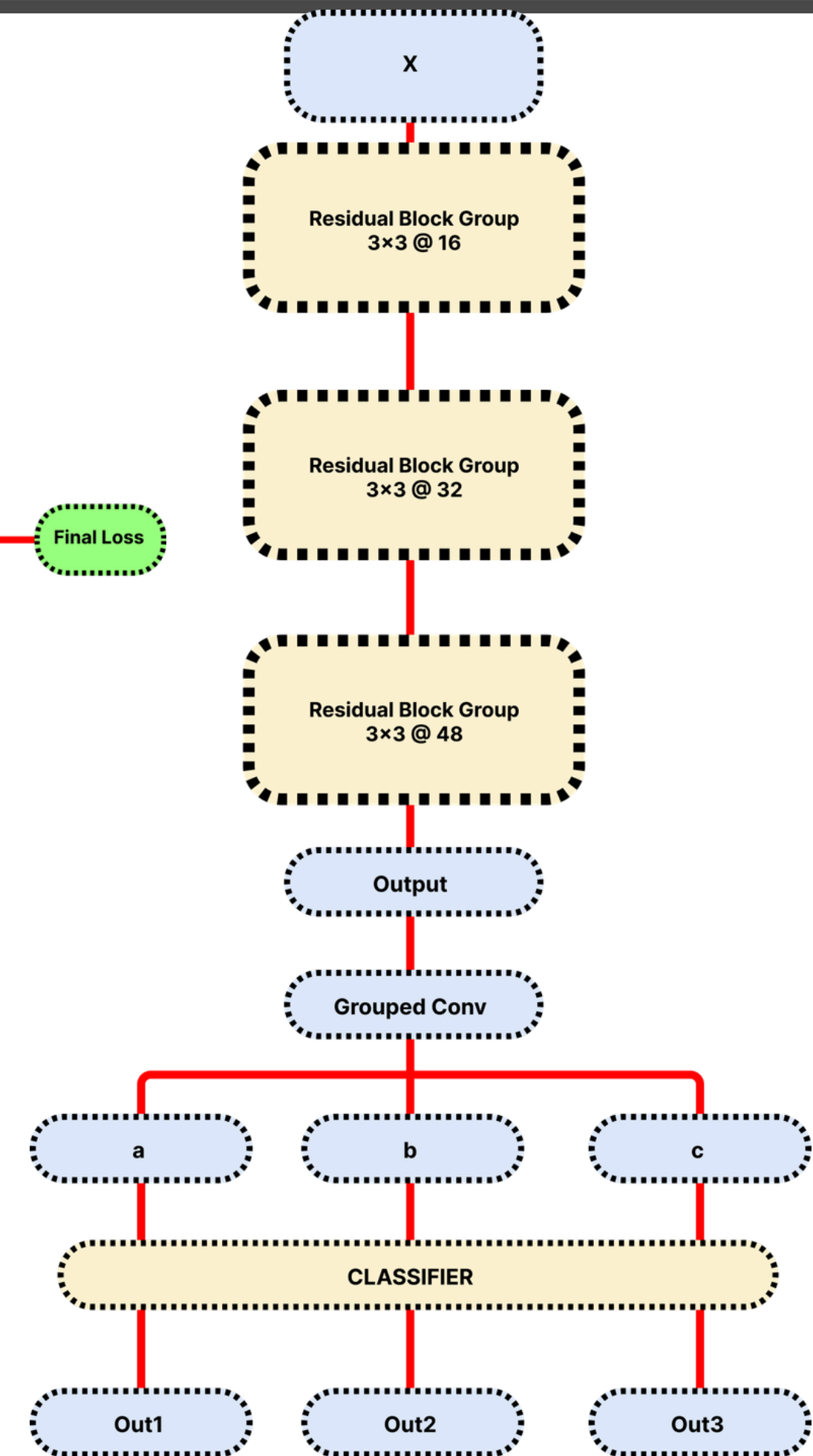
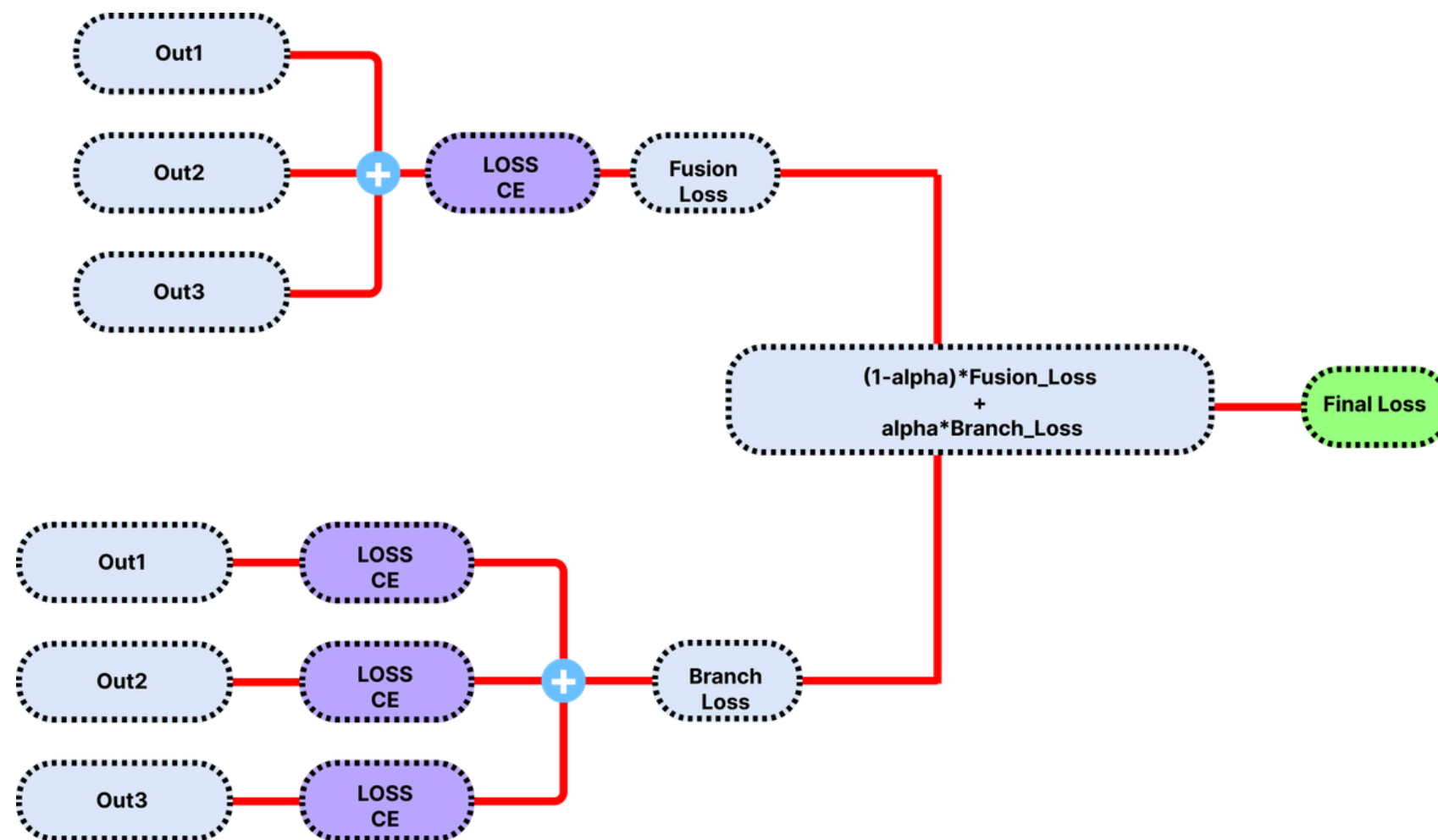
ELF's Contribution: Example-Level Hardness

- Introduced the concept of example hardness.
- Its early-exiting framework filters out "easy" images to focus the model's capacity on "hard" ones.

PHASE 1.1 - IMPLEMENTING RESLT

My first step was to create a strong baseline by implementing the ResLT paper.

- **Architecture:** I implemented a ResNet-32 backbone with the specialized ResLT head (a grouped convolution with 3 groups).
- **Practical Challenge:** I encountered a ValueError as the standard ResNet-32's 64 output channels are not divisible by 3. I resolved this by modifying the final block to output 48 channels.
- **Fusion Loss:** A standard Cross-Entropy loss on the summed output of the three branches.
- **Branch-Independent Loss:** A specialized loss calculated on individual branches using filtered data subsets to focus on medium and tail classes.



PHASE 1.2 - TUNING THE RESULT MODEL

The model's performance was critically dependent on the alpha hyperparameter. I performed extensive tuning on the 100x imbalanced dataset with different alpha vlaues to get the optimal value.

100x Models	Overall	Many	Medium	Few	
Alpha= 0.995	53.9	27.8	46.93	78.7	Tail-Specialist
Alpha= 0.9	72	83.37	66.27	67.95	Head-Specialist
Alpha= 0.99	61.77	43.9	52.5	82.12	Tail-Specialist
Alpha= 0.95	70.79	74.73	64.73	72.45	Most Balanced:

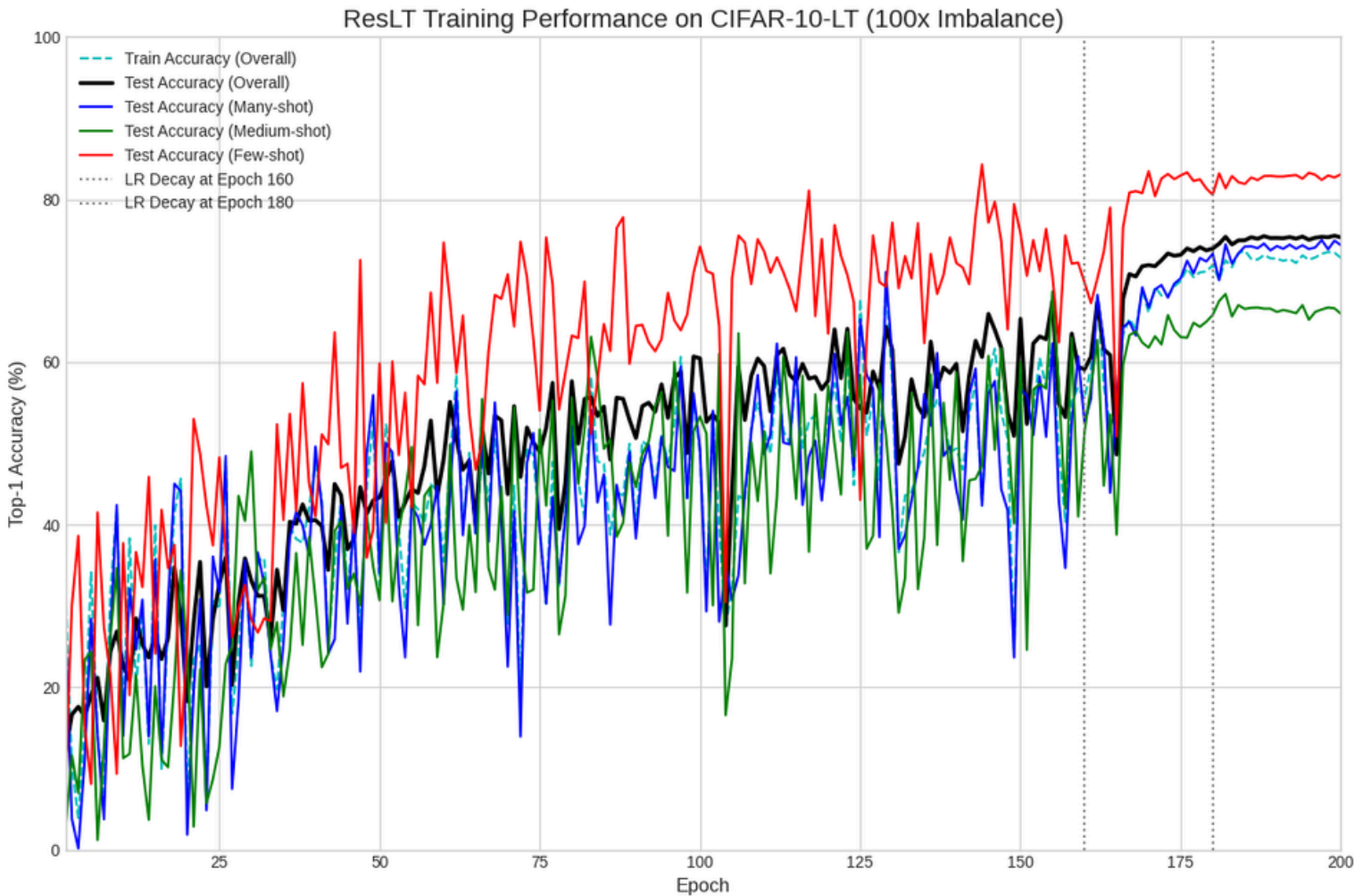
- The run with alpha = 0.95 was the most successful. It achieved a final accuracy of ~71% and, most importantly, produced a balanced model that performed well across all three class splits (Many: ~75%, Medium: ~65%, Few: ~73%). the final phase focused on closing the remaining ~9% gap to the paper's reported ~80% accuracy.

100x Models	Overall	Many	Medium	Few
Alpha= 0.95	70.79	74.73	64.73	72.45
Alpha= 0.95 with AutoAugmentation	70.22	67.87	62.3	77.92
Alpha= 0.95 with AutoAugmentation and Label Smoothing	75.33	74.37	65.93	83.1

PHASE 1.3 - FINAL RESLT MODEL

Alpha= 0.95 with AutoAugmentation and Label Smoothing	Paper's Accuracy	Overall	Many	Medium	Few	Difference
100x	80.44	75.33	74.37	65.93	83.1	~5.11
50x	83.46	80.14	80.5	70.1	87.4	~3.32
10x	89.06	86.78	84.63	79.23	94.05	~2.28

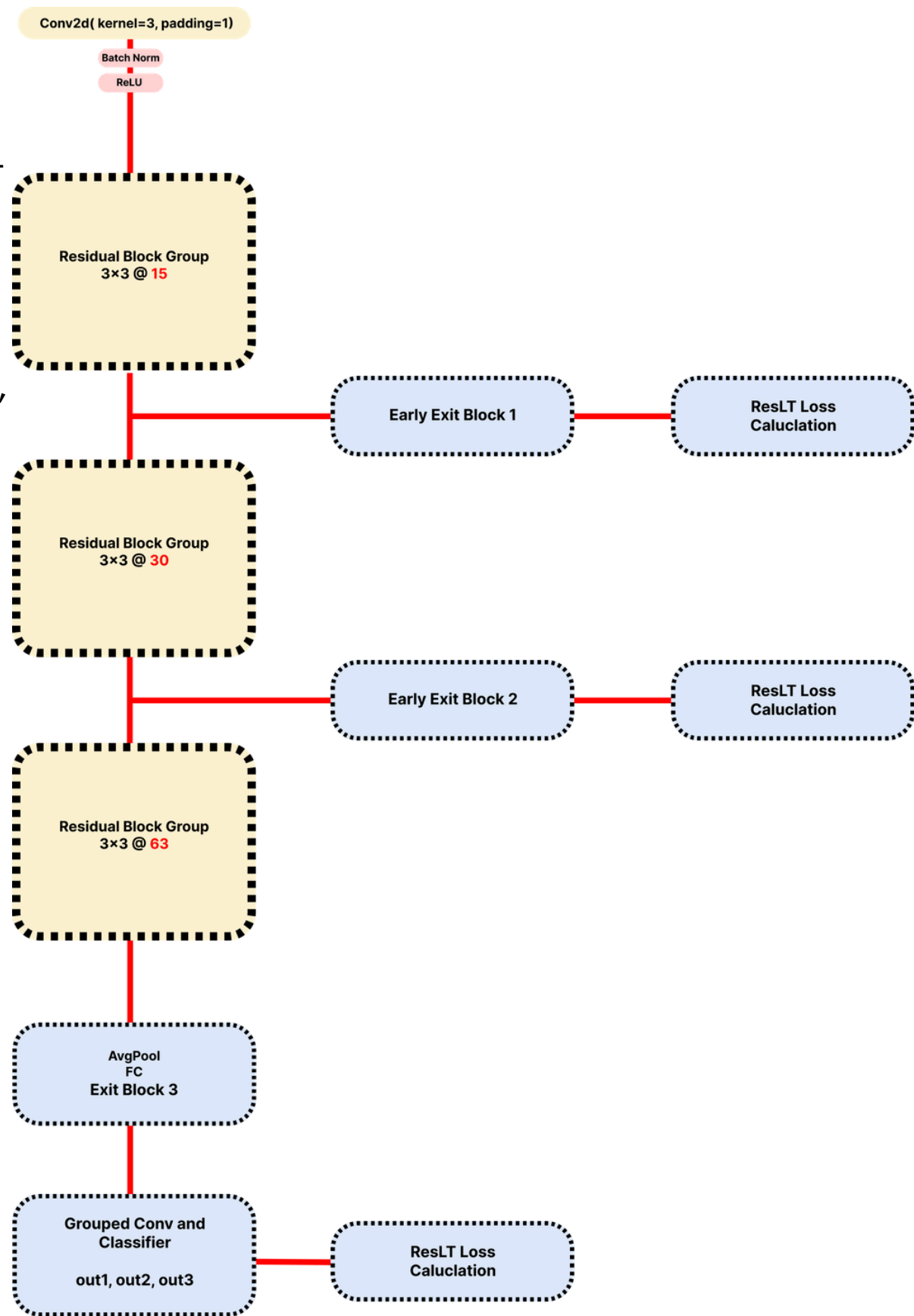
- The model with alpha value as 0.95 along with AutoAugmentation and Label Smoothing yielded me the best accuracy with a difference of -5.11%, -3.32%, -2.28% at 100x, 50x and 10x respectively.
- I moved on to the implementation of the hybrid model combining the ELF and ResLT implementation.



PHASE 2 - THE HYBRID MODEL ARCHITECTURES

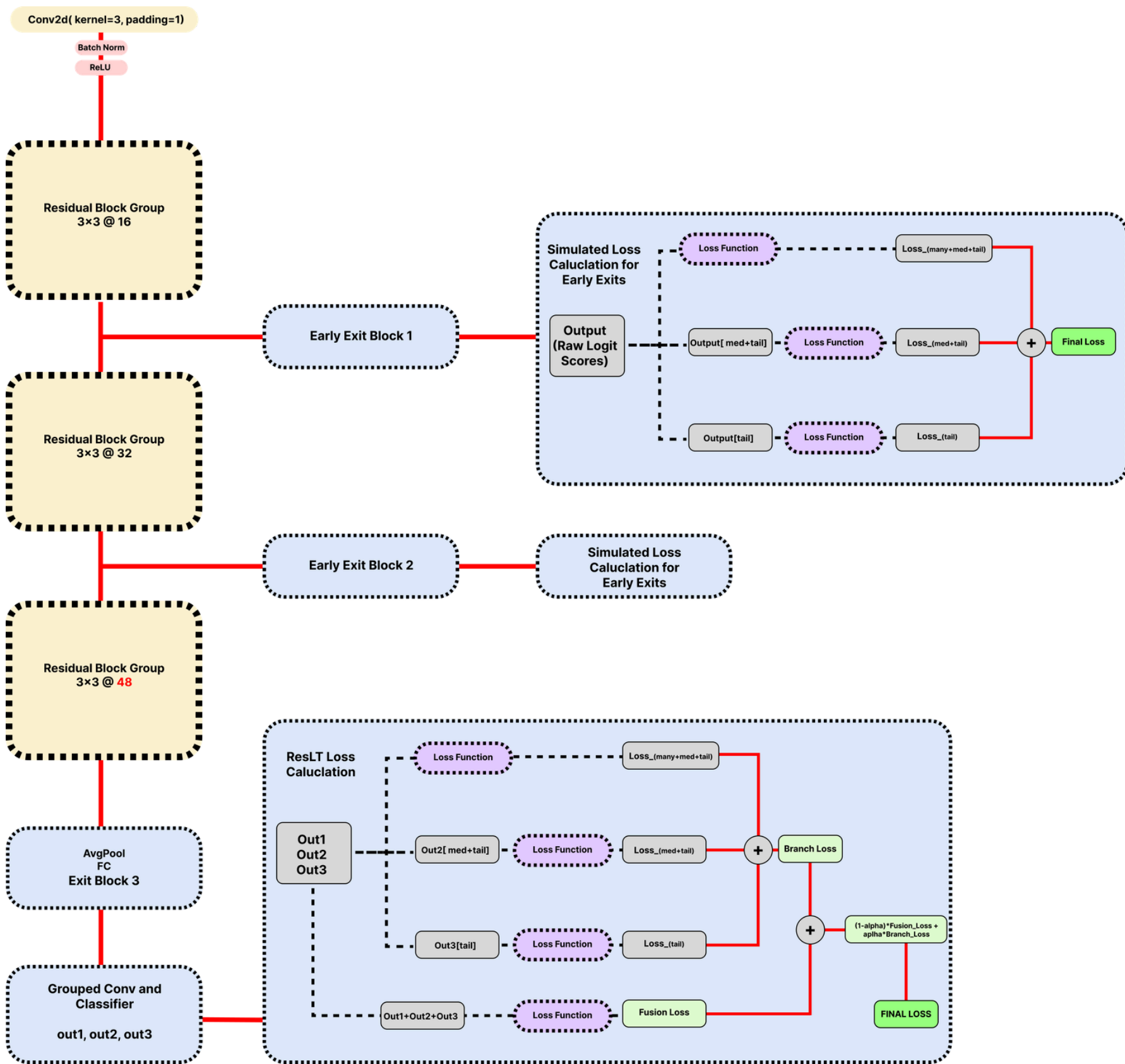
Full Power Architecture:

Include ResLT Loss calculation at every exit of the model, with ELF backbone.



Simulated Loss Architecture:

Include ResLT Loss calculation at every exit of the model, with ELF backbone.



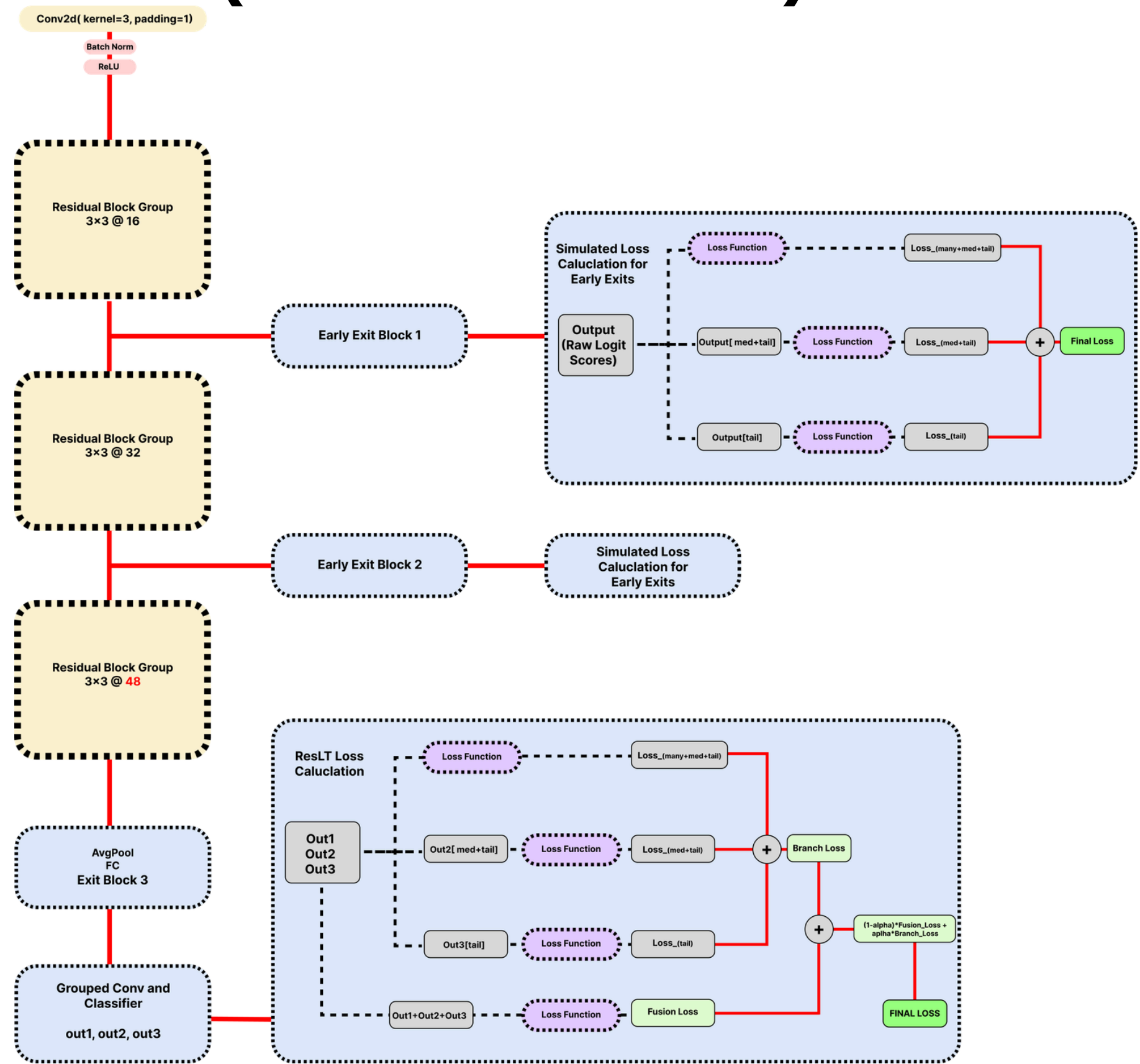
PHASE 3.1 - THE "ACCUMULATIVE HYBRID" (SIMULATED LOSS)

(MODELS 1-3)

- My initial idea was a direct combination of the two frameworks, using ELF's loss accumulation principle. Used Simulated Loss for these models for early exits alone and included a full ResLT Loss at the Final exit

100x Models	Accuracy
Model 1: Accumulative Hybrid w/ Cross-Entropy	56.9
Model 2: Accumulative Hybrid w/ LDAM, threshold=0.2	64.94
Model 3: Accumulative Hybrid w/ LDAM, threshold=0.8	68.09

- The performance was underwhelming. The accumulative loss created a noisy and difficult optimization problem, leading to a suboptimal result.
- Model 3 seems to be a further improvement, but this design seemed to have a performance ceiling well below our target. The accumulative loss was likely too complex.



PHASE 3.2 - THE "ROUTED HYBRID" (W/O LOSS ACCUMULATION)

(MODELS 4 & 7)

- Based on the instability of the first models, I did the same *but did not accumulate the losses for each sample from the previous exits*. I tried this to address the complexity of the model.

100x Models	Overall	Many	Medium	Few
Model 4: Routed Hybrid w/ LDAM	CRASHED			
Model 7: Routed Hybrid w/ CE and Label Smoothing	78.99	92.63	72.43	73.67

100x Models	Acc
Model 1: Accum	56.9
Model 2: Accum	64.94
Model 3: Accum	68.09

Model 4 (Routed Hybrid w/ LDAM)

- Design: Instead of accumulating loss, I would route each sample to a single exit for its loss calculation. This "single-point loss" was designed to be much more stable.
- Result: **CRASHED**.

Model 7 (Routed Hybrid w/ CE)

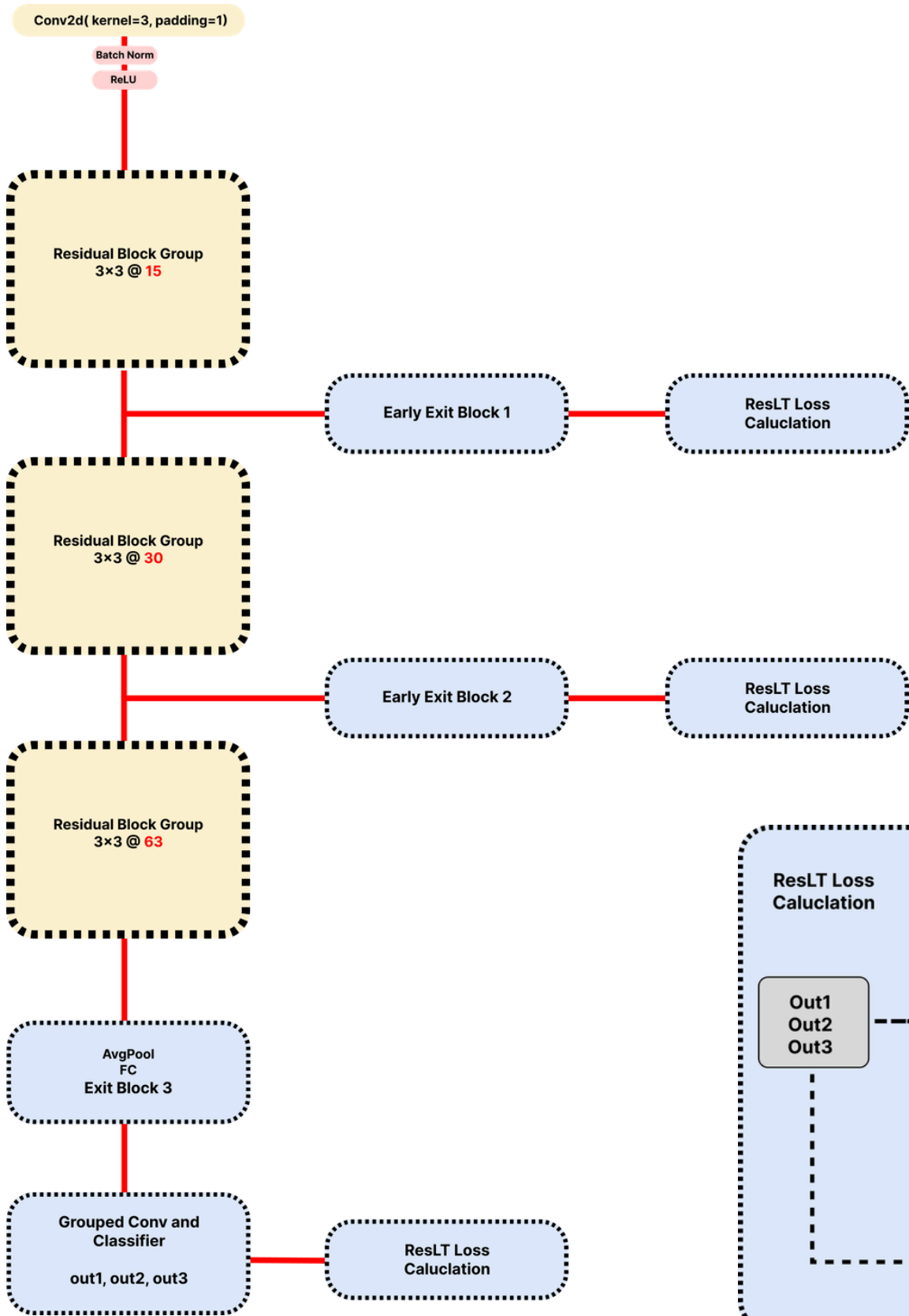
- Design:** I retried the "Routed" design, but with the more stable CrossEntropyLoss (with label smoothing) to prevent collapse.
- Result: A major success!** This model was stable and achieved a final accuracy of **78.99%**, outperforming our individual models and getting very close to the state-of-the-art.

PHASE 4.1 - THE "FULL POWER HYBRID" WITH LOSS ACCUMULATION

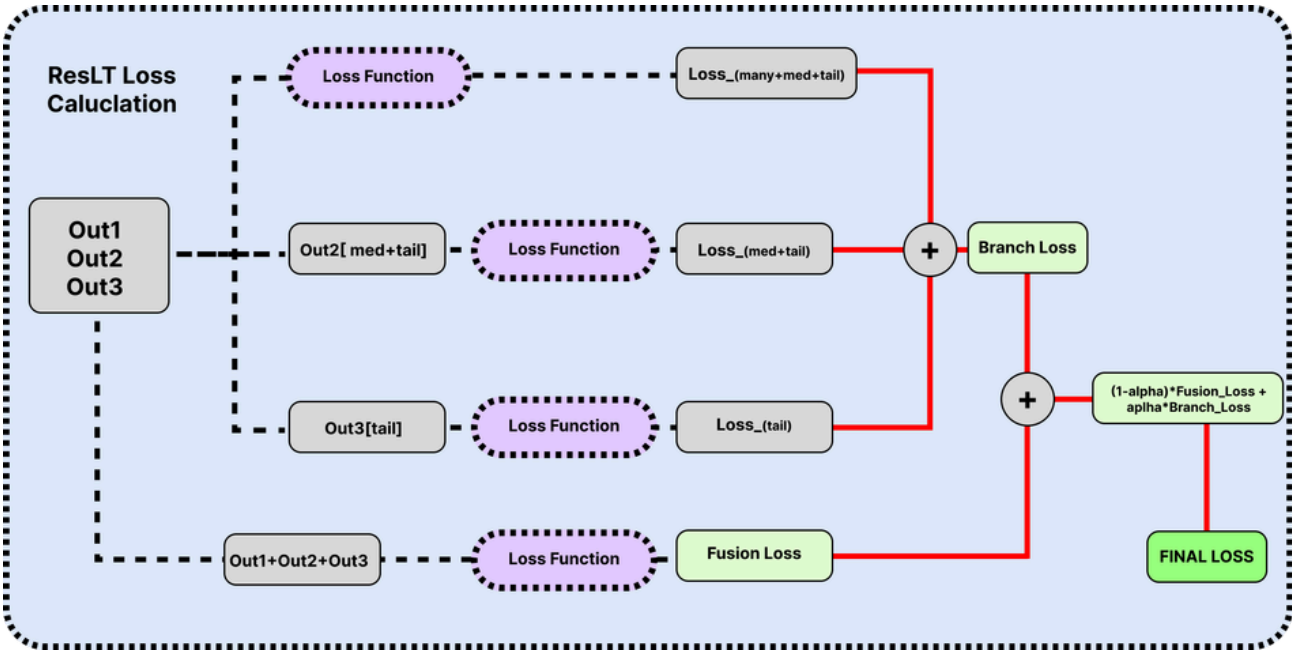
(MODELS 5,6)

- I built a model where every single exit was a full, computationally expensive ResLT head, trained with accumulative Loss at first

100x Models	Overall	Many	Medium	Few
Model 5: LDAM with Loss accumulation	60.26	29.33	57.1	85.83
Model 6: CE with Loss accumulation	62.71	79.8	56	54.8



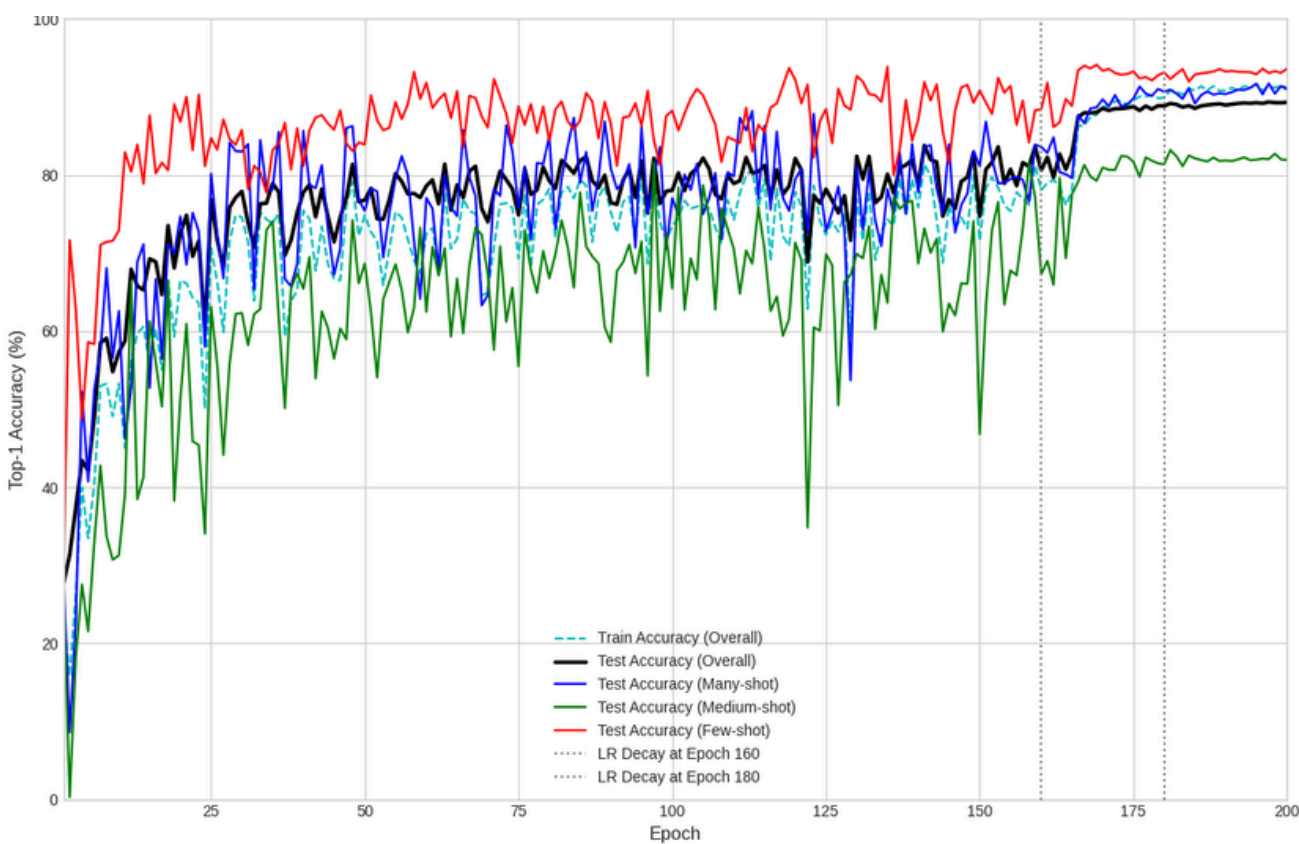
100x Models	Acc
Model 1: Accum	56.9
Model 2: Accum	64.94
Model 3: Accum	68.09
Model 4: Routed	Crashed
Model 7: Routed	78.99
Model 5: Accum	60.26
Model 6: Accum	62.71



PHASE 4.2 - THE "FULL POWER HYBRID" W/O LOSS ACCUMULATION (MODELS 14-16)

- I built a model where every single exit was a full, computationally expensive ResLT head, trained *without accumulation of loss from earlier exits*

Full Power Hybrid: Cross Entropy, AutoAugment, w/o loss accumulation	Overall Accuracy	Many	Medium	Few
Model 14: 100x	78.94	93	72.3	73.38
Model 15: 50x	83.36	92.3	75.83	82.3
Model 16: 10x	89.32	90.27	81.93	93.62



Models		Acc
Model 1:Accum		56.9
Model 2:Accum		64.94
Model 3:Accum		68.09
Model 4: Routed		Crashed
Model 7: Routed		78.99
Model 5:Accum		60.26
Model 6:Accum		62.71
Full Power: Routed	Model 14: 100x	78.94
	Model 15: 50x	83.36
	Model 16: 10x	89.32

PHASE 5 - FINAL OPTIMIZATION OF THE ROUTED MODELS (SIMULATED LOSS) (MODELS 8, 12, 13)

- Design: I took my best architecture (Model 7) and replaced AutoAugment with the more powerful Mixup augmentation on 100x.
- **Result: 79.59%.**
- A new best, this confirmed that stronger regularization was a key path to improvement.

Routed Hybrid: CE, w/o loss accumulation, Mixup Augmentation	Overall Accuracy	Many	Medium	Few
Model 8, 100x	79.59	90.4	69.93	78.72
Model 12, 50x	82.89	89.9	75.97	82.83
Model 13, 10x	85.77	83.63	76.43	94.38

Models		Acc
Model 1:Accum		56.9
Model 2:Accum		64.94
Model 3:Accum		68.09
Model 4: Routed		Crashed
Model 7: Routed		78.99
Model 5:Accum		60.26
Model 6:Accum		62.71
Full Power: Routed	Model 14: 100x	78.94
	Model 15: 50x	83.36
	Model 16: 10x	89.32
Model 7 with Mixup	Model 8, 100x	79.59
	Model 12, 50x	82.89
	Model 13, 10x	85.77

PHASE 6 - FINAL COMPARISON & RESULTS

- My final hybrid models successfully achieved state-of-the-art performance, outperforming my own strong baselines and the original papers.
- The Full Pwer Model w/o Loss Accumulation [Models 14,15,16] and Simulated Loss with Mixup Augmentation [Models 8,12,13] are the best ever model ,yielding higher accuracy

Models	100x	50x	10x
ResLT Paper’s Accuracy	80.44	83.46	89.06
ELF Paper’s Accuracy	78.1	82.4	88
My Implementation of ResLT	75.33	80.14	86.78
My Implementation of ELF	72.29	76.74	82.67
Routed Hybrid: CE, w/o loss accumulation, Mixup Augmentation. Models[8,12,13]	79.59	82.89	85.77
Full Power Hybrid: Cross Entropy, AutoAugment, w/o loss accumulation. Models[14,15,16]	78.94	83.36	89.32

Models		Acc
Model 1:Accum		56.9
Model 2:Accum		64.94
Model 3:Accum		68.09
Model 4: Routed		Crashed
Model 7: Routed		78.99
Model 5:Accum		60.26
Model 6:Accum		62.71
Full Power: Routed	Model 14: 100x	78.94
	Model 15: 50x	83.36
	Model 16: 10x	89.32
Model 7: Routed, with Mixup	Model 8, 100x	79.59
	Model 12, 50x	82.89
	Model 13, 10x	85.77

CONCLUSION

Key Findings:

- "Routing" Loss is More Stable than Accumulating Loss and boosts accuracy as well
- The Full Power Architecture, despite being computationally expensive, yields higher accuracy than all the other models
- The Simulated Loss struggles to achieve paper's level and is boosted by Mixup Augmentation only. However Full Power achieved higher accuracy with AutoAugment
- The Simulated Model performs better at 100x model than Full Power Model.
- The Full Power Model performs better at 10x dataset than Simulated Model

	Models		Acc	Difference
Loss Accumulated	Model 1:Accumulative Hybrid w/ Cross-Entropy		56.9	-23.54
	Model 2:Accumulative Hybrid w/ LDAM, threshold=0.2		64.94	-15.5
	Model 3:Accumulative Hybrid w/ LDAM, threshold=0.8		68.09	-12.35
Loss not Accumulated	Model 4: Routed Hybrid w/ LDAM		Crashed	-
	Model 7: Routed Hybrid w/ CE and Label Smoothing		78.99	-1.45
Loss Accumulated	Model 5: Full Power: LDAM with Loss accumulation		60.26	-20.18
	Model 6: Full Power: CE with Loss accumulation		62.71	-17.73
Loss not Accumulated	Full Power Hybrid: Cross Entropy, AutoAugment, w/o loss accumulation	Model 14: 100x	78.94	-1.5
		Model 15: 50x	83.36	-0.1
		Model 16: 10x	89.32	/+0.26
Loss not Accumulated	Model 7: Routed Hybrid w/ CE and Label Smoothing, with Mixup	Model 8, 100x	79.59	-0.85
		Model 12, 50x	82.89	-0.57
		Model 13, 10x	85.77	-3.29